

Hands-On Case Studies in JavaScript (Module1-3)

Case Study 1: Student Grade Management System

Scenario

You are tasked to build a simple **Grade Management System** using JavaScript. The system should:

1. Ask the user for the student’s **name**, **age**, and **course**.
2. Ask for **3 grades** and store them in an array.
3. Compute the **average grade**.
4. Classify the student’s status:
 - o "Passed" if average ≥ 75
 - o "Failed" if average < 75
5. Display the student’s info, grades, average, and status using alert() or console.log().

Instructions

1. Create an index.html and link a main.js file.
2. In main.js, implement the Grade Management System.
3. Test your program with different grades.

Expected Output (Example)

Student Information:
Name: Juan Dela Cruz
Age: 20
Course: IT
Grades: 80, 85, 90
Average: 85.00
Status: Passed

Learning Outcomes

- Using **variables** and **objects** for student info.
- Using **arrays** for storing grades.
- **Loops** for grade processing.
- **Conditionals** for status classification.
- **Dialogs** for user interaction.

Case Study 2: Simple ATM Simulation

You are tasked to create a **Simple ATM Program** using JavaScript. The ATM should:

1. Ask the user to enter an **initial balance**.
2. Display a **menu** of options: Deposit, Withdraw, Check Balance, Exit.
3. Perform the transaction based on user choice.
4. Prevent withdrawal if the amount is greater than the balance.
5. Continue showing the menu until the user chooses Exit.

Instructions

1. Create an index.html and link a main.js file.
2. Implement the ATM program in main.js.
3. Test your program with different transactions.

Expected Output (Example)

You deposited 500. New balance: 1500

You withdrew 300. New balance: 1200

Your current balance is: 1200

Thank you for using the ATM. Goodbye!

Learning Outcomes

- **Loops** (do...while) to create a repeating menu.
- **Conditionals** for menu logic.
- **Arithmetic operations** for transactions.
- **Validation** for sufficient funds.
- **Dialogs** (prompt, alert) for user interaction.